[1993AstL..19..133D]

Blurring of spatial microwave background fluctuations by molecular line scattering

V. K. Dubrovich

Special Astrophysical Observatory, Russian Academy of Sciences, Nizhnii Arkhyz

(Submitted March 24, 1992; resubmitted September 1, 1992)

Pis'ma Astron. Zh. **19**, 132–135 (February 1993)

Molecular scattering of radiation in the expanding universe at redshifts z from 30 to 50 is considered. It is shown that the presence of molecules like H_2D^+ can create a "curtain" that blurs the primordial spatial fluctuations of the cosmic background radiation (CBR). The required abundances for this molecule are estimated and features of the secondary fluctuations are discussed.

One of the most popular problems of modern observational radio astronomy is the search for primordial spatial fluctuations in the background radiation (BR). Their existence is due to the presence of primordial mass-density fluctuations which were the cause of the present galactic clusters. The most intense BR fluctuations formed when the universe became transparent to radiation, and therefore carry information about mass-density fluctuation amplitudes and length scales at $z = 1000$ (Zel'dovich and Novikov, 1975). Unfortunately, agreement between theoretical predictions and observational data is not very good.

One way of saving the overall scheme is the secondary warming hypothesis for matter at small z. Here the complete ionization of hydrogen is possible, with the return of opacity. However, detailed analysis of this version does not provide confidence in its adequacy, because the actual mass-density and the Thomson cross section for free electron scattering are too small. Furthermore, one must find a sufficiently strong energy source for the total ionization of hydrogen. In the present paper we consider the possibility of opacity due to scattering by primordial molecules.

Molecular scattering differs significantly from free electron scattering. Firstly, it only occurs in narrow spectral lines, and secondarily, the effective cross section is greater by 5-6 orders of magnitude than the Thomson cross section, i.e., there will be a noticeable effect even when the molecular populations are small. Line scattering leads to the fact that the layer of matter found at distances $z_2 < z < z_1$ will be opaque (for a present-day observer) in the wavelength interval from $\lambda_1 = \lambda(1 + z_1)$ to $\lambda_2 = \lambda(1 + z_2)$, where λ is the laboratory wavelength of the molecular transition and z_1, z_2 are respectively the far and near boundaries of the scattering layer. The spectrum of molecular transitions contains many lines. In practice the opacity of the layer under consideration will occur simultaneously in several bands. If the width of the layer is sufficiently great, i.e., the molecules are created at z_1 and disappear at z_2 such that $z_2 < 0.5z_1$, then photons will be scattered at all wavelengths, with no gaps, starting at some maximum wavelength $\lambda_{10} = \lambda_0(1 + z_1)$, where λ_0 is the wavelength of the first rotational transition. The short-wavelength opacity boundary depends on the optical depth in vibrational lines. The ratio $z_2/z_1 = 2$ is given by the frequency ratio of the first and second rotational transitions in diatomic molecules. The relative distance between the later lines is less than 2.

We now consider the question of secondary fluctuations. The fluctuation formation mechanism at this time is exactly the same as at the time of hydrogen recombination. However, the nature of the fluctuations is completely different. Indeed, the homogeneous and isotropic matter background uniformly washes out the primordial fluctuations, independently of the scattering mechanism. Density fluctuations can be regarded as objects of finite size with peculiar velocities (relative to the frame of reference in which the BR is isotropic). But, being opaque only in the lines, these objects will only be visible in a narrow frequency band governed, as was shown above, by their geometrical dimensions (Dubrovich, 1982). With free electron scattering they would have been visible at all frequencies. The spectrum of angular scales for secondary fluctuations is shifted into the small scale domain in proportion to $(z_2/1000)^{3/2}$. This is a qualitative picture of radiation scattering by matter containing arbitrary molecules, taking into account the expansion of the universe. If we are interested in the possibility of using these scattering properties to explain the lack of BR fluctuations, then we have to ensure, firstly, that the observable wavelength range from 10-15 cm to 1-2 mm is covered, and secondly, that the optical depth $\tau > 1$ throughout that range. From the expression for the maximum wavelength, it follows that it is necessary to select molecules which either have transitions of sufficiently low frequency in their spectra, or which are formed at large z. The latter condition is equivalent to the requirement that the molecule have a large dissociation potential.

We now consider specific molecules. The primordial pulsations of chemical elements sharply diminish with increasing atomic number, so one is restricted to hydrogen deuterium, helium, and perhaps lithium. To a first approximation they will form diatomic molecules. Analysis of all combinations of these elements taking account of the above requirements leads to a negative conclusion (Dubrovich, 1977). One cannot simultaneously cover the necessary frequency range and obtain large optical depth. Combinations of different molecules do not help either. Hence we shall consider the possibility of using triatomic molecules. It turns out that interesting results are obtained in the case of H_2D^+. The HD^+ molecule is rapidly destroyed by collisions with neutral hydrogen atoms, while H_2D^+ is stable under such collisions. But its main merit is the presence of low-frequency transitions, high dissociation potential, and a significant deuterium population which is greater by 4-5 orders of magnitude than that of lithium. It is true that the

 1063-7737/93/01 0053-02 $10.00 © 1993 American Institute of Physics

© American Institute of Physics • Provided by the NASA Astrophysics Data System

[1993AstL..19...53D]

formation kinetics of this molecule are relatively complex. Thus, the basic mechanism governing the H_2D^+ population is the deuterization of H_3^+. In view of the rapid collisional reactions rates, the ratio of the $[H_2D^+]$ and $[H_3^+]$ concentrations will be proportional to the ratio of the [D] and [H] concentrations, essentially as in thermodynamic equilibrium:

$$[H_2D^+]/[H_3^+] \approx [D]/[H] \exp(T_E/T_C), \quad (1)$$

where T_E, the difference between the dissociation energies of these molecules, is ~ 300 K when expressed as a temperature (Adams and Smith, 1981), T_C is the kinetic temperature of the matter, which at $z < 150$ can be significantly lower than the temperature of the BR (T_r); square brackets denote concentrations of the respective molecules. As the temperature falls, the concentration of H_2D^+ rises and in the limit is governed by the complete capture of all the deuterium, i.e., the upper limit on the H_2D^+ population is $2 \cdot 10^{-5}$ that of hydrogen. The matter temperature required for this occurs for z between 50 and 30. An interesting feature of this process is the strong dependence of the H_2D^+ population on T_C. Inside denser regions, T_C is always greater than in the surrounding medium, because of warming due to the contraction of proto-objects and because of fraction during their motion with some peculiar velocity. Thus the fluctuations in any case become transparent earlier than the surrounding stationary medium and the secondary BR fluctuations they cause wash themselves out, which significantly diminishes their magnitude. The main mechanism of molecular disruption is recombination with electrons.

We estimate the optical depth τ in the molecular lines taking the expansion of the universe into account:

$$\tau = \left[1 - \exp\left(-\frac{h\nu}{kT_r}\right)\right] \int \sigma(\nu) N_i(z) \frac{c\,dt}{dz}\, dz$$
$$= \left[1 - \exp\left(-\frac{h\nu}{kT_r}\right)\right] \frac{8\pi^2 S}{h} N_i \frac{c\,dt}{dz} z. \quad (2)$$

where h is Planck's constant, ν is the frequency of the scattered photon, k is Boltzmann's constant, T_r is the radiation temperature, S is the line strength, N_i is the molecular concentration in the ith excited state, from which photons of frequency ν are absorbed, and t is the cosmological time. Putting in numerical values, we obtain

$$\tau \sim 4.5 \cdot 10^7 \left[1 - \exp\left(-\frac{h\nu}{kT_r}\right)\right] \alpha S_d \left(\frac{N_B}{N_C}\right) \left(\frac{H_0}{75 \frac{\text{km}}{\text{sec} \cdot \text{Mpc}}}\right) \left(\frac{z}{100}\right)^{3/2},$$
$$\alpha = \frac{N_i}{N_B}, \quad N_C = \frac{3H_0^2}{8\pi G m_p}. \quad (3)$$

where S_d is the line strength in units of D^2. Taking the line parameters to be (Khersonskii et al., 1987) $\nu_1 = 12.43$ cm^{-1}, $S_d = 0.54$, $E_1 = 86.4$ K, and $\nu_2 = 5.203$ cm^{-1}, $S_d = 1.13$, $E_1 = 314.5$ K, where E_1 is the excitation energy of the sublevel from which the scattering of the given line takes place, and substituting these quantities into the expression for τ, we find that the second transition will scatter radiation starting at $\lambda = 10$ cm (at $z = 50$), and $\tau = 1$ at $\alpha = 10^{-6}$. Such a concentration of H_2D^+ is only possible in the presence of a sufficiently high concentration of free electrons, of the order of 10^{-3}, i.e., in the presence of a source of high-energy photons. Such a source could be the break-up of unstable particles. However, in this case it is necessary to have photons that can ionize not only atoms, but also molecules of hydrogen, i.e., with energies > 15.5 eV.

A more thorough consideration of this mechanism for blurring primordial fluctuations does, of course, require detailed calculations of the molecule formation kinetics, determination of the populations of the excited levels, the role of secondary ionization sources, and matter warming. There is a separate question concerning the role of H_2D^+ in the cooling of primordial proto-objects. In any case, analysis of consequences of the existence of triatomic molecules in the early universe is of great importance.

This work was supported by grants from the Council for Cosmological Microphysics and the American Astronomical Society.

Adams, N. G. and Smith, D. (1981). *Astrophys. J.* **248**. 373.

Dubrovich, V. K. (1977). *Pis'ma Astron. Zh.* **3**, 243 [*Sov. Astron. Lett.* **3**, 128 (1977)].

Dubrovich, V. K. (1982). *Izv. Spets. Astrofiz. Observ. Zh.* **15**, 21.

Khersonskii, V. K., Varshalovich, D. A., and Opendak, M. G. (1987). *Astron. Zh.* **64**, 520 [*Sov. Astron.* **31**, 274 (1987)].

Zel'dovich, Ya. B. and Novikov, I. D. (1975). *Structure and Evolution of the Universe* [in Russian], Nauka, Moscow.

Translated by R. L. Znajek

© American Institute of Physics • Provided by the NASA Astrophysics Data System